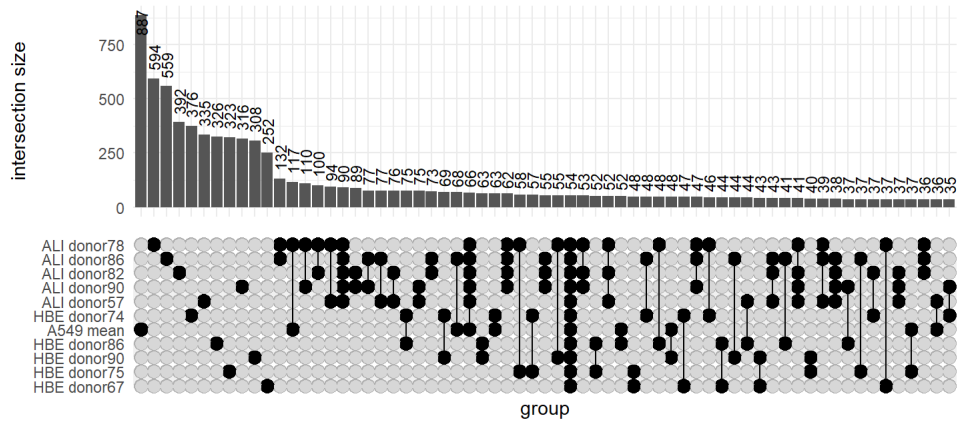
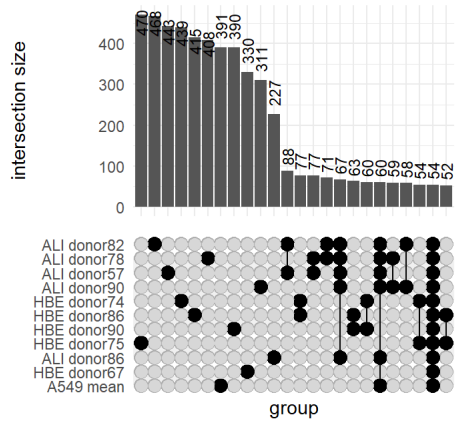
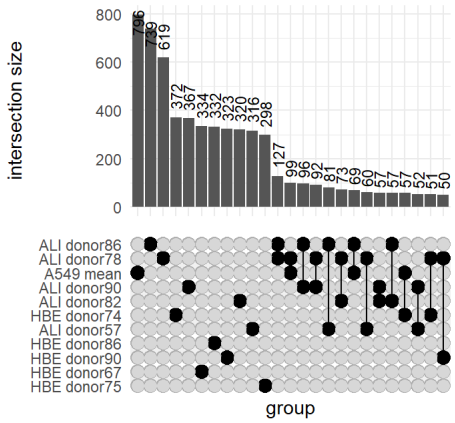
(A) IL1B-regulated



(B) IL1B-upregulated



(C) IL1B-downregulated



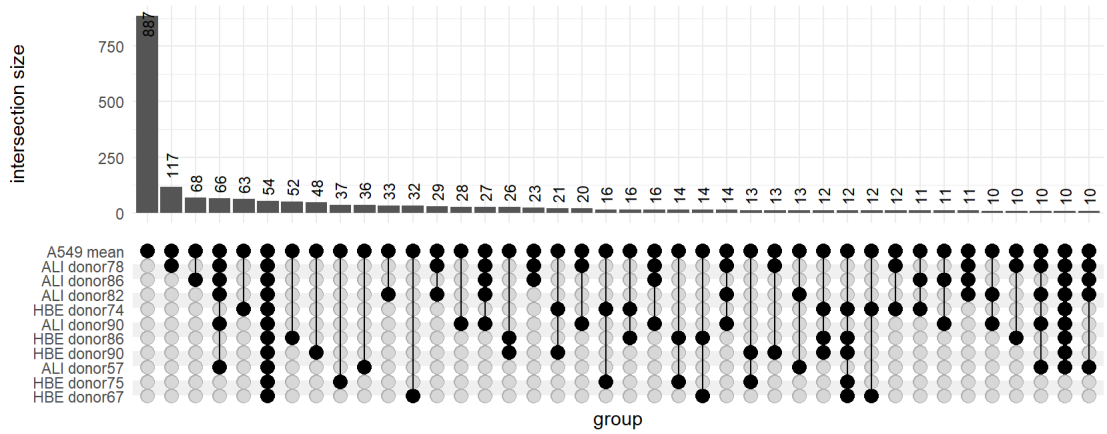
2.3. IN PROGRESS: A549s effectively capture what “could” happen across donors

Findings from 2.2. suggest that we should pivot from “intersection biology” to “union biology.” i.e., A549s do not recapitulate any single primary IL1B response, but capture a large fraction of the population-level transcriptional response (with a preference for induced programs). **To explore the extent to which A549-regulated gene expression overlaps with the union of all donor-regulated gene expression, fold values from the four A549 replicates were averaged into an “A549 mean” – representative of the A549 donor, to be compared against all other donors.**

(A) Showing all intersections with more than 10 genes, 887 genes were unique to A549 cells while 1040 were shared between A549 cells and at least one donor. Thus, **54% of the IL1B-regulated response in A549 cells represents something that could be observed within a population of donors.**

(B) Narrowing down into IL1B-induced genes, we see 391 unique to A549 cells anywhere between 286 (minimum intersection size = 10), and 514 shared (minimum intersection size 3). So more than 42-56% of the IL1B-induced signature in A549s is shared with at least one primary cell. **(C)** ~50% for downregulated genes.

title here



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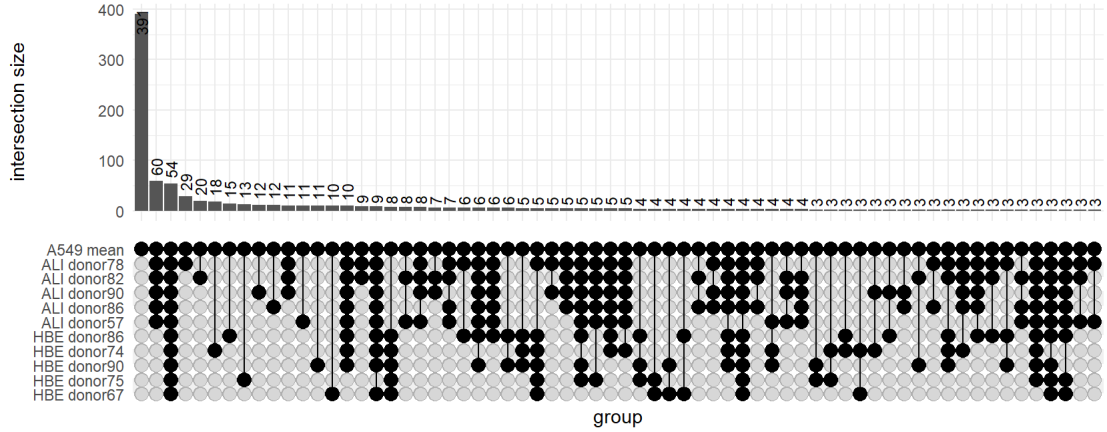
intersection size
400
300
200
100
0
391
60
54
29
20
18
15
13
12
12
11
11
11
10
10
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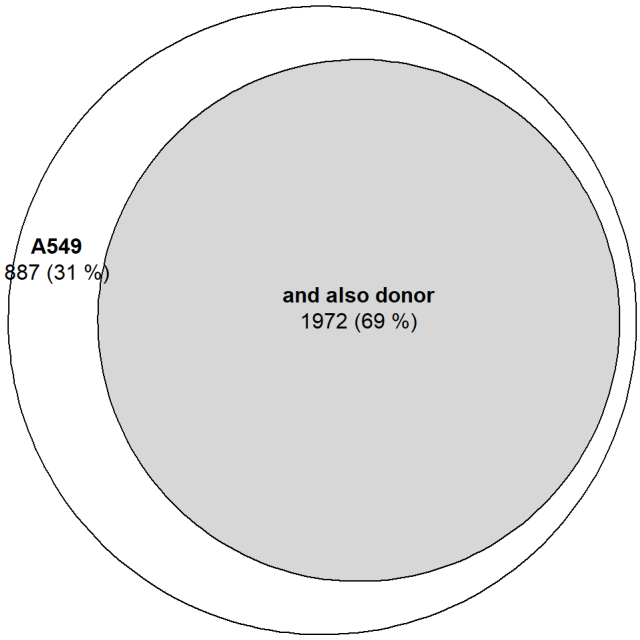
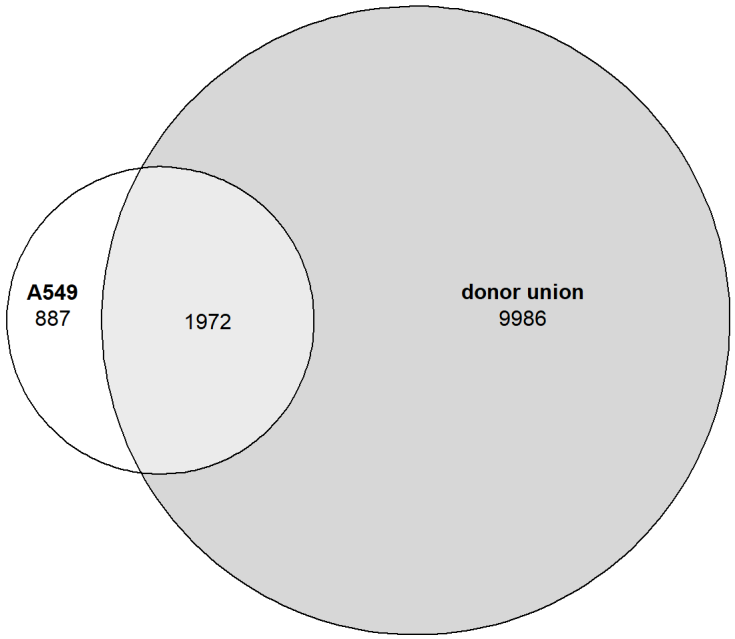


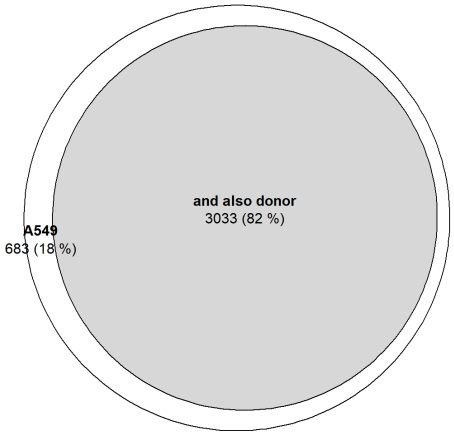
However, A549s only capture 16% of the entire "donor universe"… So while the things that happen in A549s are most likely also found in 1+ donors, A549s themselves are not a perfect system for capturing everything that could happen?? Suggests considerable individual variation… But this variation should be in peripheral/context-dependent programs, while the core NFkappaB program should be faithfully captured by A549s (and map to primaries)

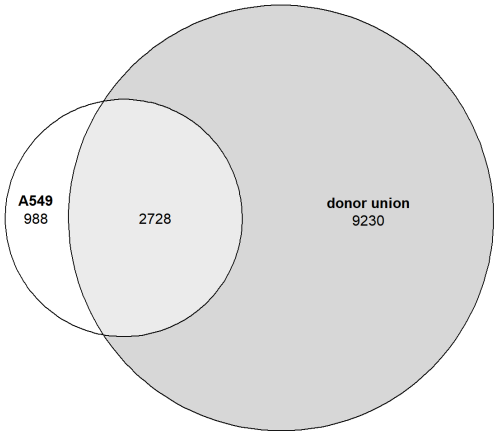
Isn't this the problem we had at the start ???



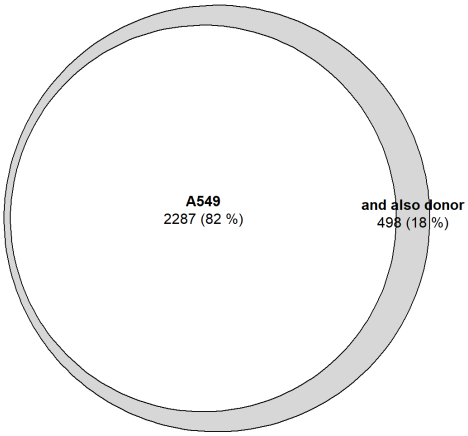
What if we take the A549 union instead of the A549 mean…

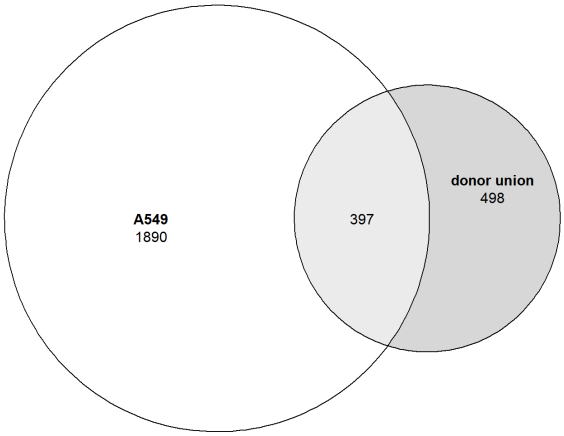
82% of A549-regulated genes is something that could happen in primaries, but only 23% of the donor universe is captured…



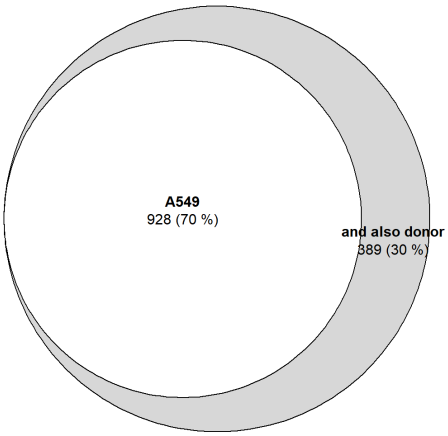


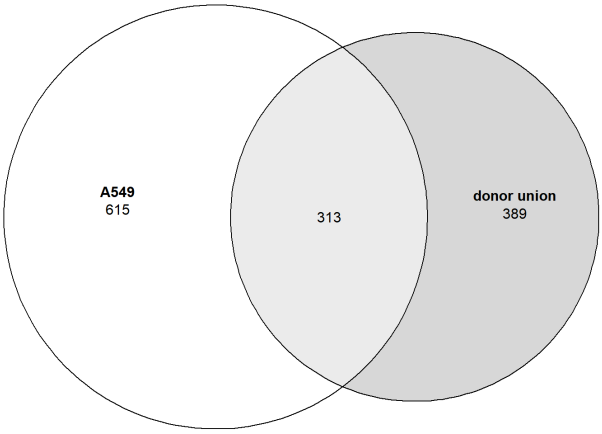
What if we do the above analysis with sleuth results…





What if we do the above with sleuth results, but only look at upregulated?





2.4. Separately, all celltypes share a small, conserved, “core” program

These genes have … functional signature, follow … kinetic, are … NFkappaB dependent (and should be the focus of future studies?)

3. **=== Figures and code to be sorted into story notes ===**